

Day 3 Exercises: Problem Solving Sessions 1 and 2

This handout combines both Day 3 problem-solving sessions.

Session 1

This session builds accuracy in iterative matrix updates and interpretation of transition entries. Focus on connecting calculations to real process behavior.

1. For $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{x}_0 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, compute $\mathbf{x}_1$.
2. Continue Problem 1 to compute $\mathbf{x}_2$.
3. For $A = \begin{bmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{bmatrix}$ and $\mathbf{x}_0 = \begin{bmatrix} 100 \\ 50 \end{bmatrix}$, compute one step.
4. Explain one sentence interpretation of entry 0.2 in row 2, column 1.
5. Why does iteration need an initial state?
6. Give one non-math real-world process that is iterative.
7. If two consecutive states become almost equal, what might that suggest?
8. In a two-state model, why should states stay nonnegative?
9. Let $A = \begin{bmatrix} 0.6 & 0.4 \\ 0.1 & 0.9 \end{bmatrix}$ and $\mathbf{x}_0 = \begin{bmatrix} 120 \\ 80 \end{bmatrix}$. Compute $\mathbf{x}_1$ and $\mathbf{x}_2$.
10. Reflection: give one model assumption that could make long-term predictions unreliable.

Session 2

This session focuses on identifying eigenvectors and eigenvalues and linking them to long-term behavior. Use the eigenvector test carefully in each problem.

1. Check whether $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is an eigenvector of $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$.
2. Check whether $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector of $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$.
3. In your own words, define eigenvalue.
4. Why are eigenvalues useful in long-term prediction?
5. Give one assumption in a two-state transition model.
6. For $A = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$, find eigenvalue for $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.
7. For the same A , what happens to the first component after many iterations if that direction is present?
8. Suppose an eigenvalue is 0.6. What does that suggest about repeated updates in that direction?
9. Reflection: explain why checking $A\mathbf{v}$ against a multiple of $\mathbf{v}$ is the key test.